Kevin Joseph Donaldson, PhD

Address: Providence, RI, USA
ORCID: 0000-0002-9144-3776
Email: kjdonaldson.contact@gmail.com
Phone: (401) 400-1403

EDUCATION

2025 **Doctor of Philosophy, Neuroscience**
Georgia State University, Atlanta, GA
Advisor: Daniel Cox, PhD

2021 **Master of Science, Neuroscience**
Georgia State University, Atlanta, GA

2015 **Master of Science, Neuroscience and Cognitive Science**
University of Maryland, College Park, MD
- Originally PhD student, advanced to candidacy 2010
- Transitioned to Master's degree 2014

2002 **Bachelor of Science, Animal Sciences**
University of Maryland, College Park, MD

RESEARCH EXPERIENCE

2018-2025 *PhD Student*, Neuroscience Institute, Georgia State University
- Characterized previously undescribed cellular and molecular mechanisms of experience-dependent cold nociceptive plasticity in *Drosophila* using super-resolution microscopy, electrophysiology, calcium imaging, and optogenetic circuit manipulation
- Contributed to the discovery of a neural basis for cold acclimation in *Drosophila*
- Developed high-throughput, open-source, optical stimulation device (Optobox) for optogenetic control of neural circuits during development

2017-2018 *Research Technical Specialist*, Emory Eye Center, Ophthalmology Department, Biostatistics and Bioinformatics (Bioanalysis) Core
- Trained Emory Eye Center (EEC) research staff and students in confocal microscopy, IMARIS 3D image analysis software, and mouse *in vivo* ophthalmic imaging techniques
- Wrote custom analysis software enabling auto-segmentation of 2D and 3D retinal images
- Assisted EEC researchers with bioinformatics and gene network analyses
- Co-supervised and maintained the EEC Imaging Core confocal microscope

2015-2017, 2019-2025 *Research Specialist*, Nickerson Lab, Department of Ophthalmology, Emory University School of Medicine
- Generated novel transgenic mouse lines enabling longitudinal, *in vivo* imaging of retinal pigment epithelium (RPE) in mouse models of retinal degeneration
- Examined the role of RPE dysmorphology in multiple mouse models of blinding disorders using immunohistochemistry and confocal microscopy

2006-2015 *Graduate Student*, Neuroscience and Cognitive Science Program, University of Maryland, College Park (UMCP)
- Investigated the role of orbitofrontal cortex in auditory perception using a novel head-fixed, behaving mouse preparation and single-unit *in vivo* electrophysiology
- Used antero- and retrograde tracers to map connections between frontal and sensory cortices in mice and rats

- Mapped the neuronal network involved in auditory task performance using c-fos immunohistochemistry
- Designed and conducted auditory perception experiments utilizing human MEG physiology in collaboration with UMCP Linguistics Department

2005-2006 *Faculty Research Assistant*, Neural Systems Lab, UMCP
- Primary behavioral trainer of ferrets used in operant auditory discrimination tasks
- Organized and maintained lab resources for behavioral training and physiology experiments

2002-2005 *Faculty Research Assistant*, Halperin Animal Behavior Lab, UMCP
- Conducted behavioral experiments using *Betta splendens* model to test the role of social isolation in aggression
- Trained and supervised a staff of 8 undergraduate researchers
- Constructed custom fish tanks enabling habitat-like conditions for behavioral experiments

2002 *Statistical Analyst*, Christman Biometrics Lab, UMCP
- Acquired juvenile fin fish from Maryland's Chesapeake Bay and recorded biometric measurements
- Combined biometric data with geographic information systems (GIS) data as part of a statewide survey of juvenile fish health and stability in the Chesapeake Bay for the Department of Natural Resources

2001-2002 *Research Assistant*, Carr Auditory Neurophysiology Lab, UMCP
- Maintained lab equipment, solutions, and experimental animals (barn owls)
- Prepared auditory physiology experiments investigating sound localization
- Constructed pre-experiment isolation chamber and enrichment enclosure

PUBLICATIONS

PUBLISHED

* indicates co-first author

Donaldson KJ, Chrenek MA, Boatright JH, Nickerson JN. High resolution imaging and interpretation of three-dimensional RPE sheet structure. Biomolecules. 2025 Jul 26;15(8):1084. DOI: 10.3390/biom15081084. PMID: 40867529;

Girardot P, Zhang X, Zhang N, **Donaldson K**, Chrenek M, Sellers J, Feola A, Papania J, Nickerson J, Jain N, Boatright J. Pentosan Polysulfate Sodium Causes Diminished Function and Subtle Morphological Changes in Retina and RPE of Mice. Investigative Ophthalmology & Vision Science. Feb 2024; 65(28). DOI: 10.1167/iovs.0.0.38631. PMID: 38381414

Kirchner MK, Althammer F, **Donaldson KJ**, Cox DN, Stern JE. Changes in neuropeptide large dense core vesicle trafficking dynamics contribute to adaptive neuronal responses to a systemic homeostatic challenge. iScience. 2023 Oct 18. 26(11):108243. PMID: 38026155

Himmel N*, Sakurai A*, **Donaldson KJ*.** Protocols for measuring cold-evoked neural activity and cold tolerance in Drosophila larvae following fictive cold acclimation. STAR Protocols (Cell Press). 2022 Jun 28; 3(3). PMID: 35776643

Ferdous S, Liao KL, Gefke ID, Summers VR, Wu W, **Donaldson KJ**, Kim YK, Sellers J, Dixon J, Shelton D, Markand S, Kim SM, Zhang N, Boatright JH, Nickerson JM. Age related retinal changes in wild type C57BL/6J mice between 2 and 32 months. Investigative Ophthalmology and Visual Science. 2021 Jun 1; 62(7):9. PMID: 34100889

Himmel NJ, Letcher JM, Sakurai A, Gray TR, Benson MN, **Donaldson KJ**, Cox DN. Identification of a neural basis for cold acclimation in *Drosophila* larvae. iScience. 2021 May 28; 24(6):102657. PMID: 34151240

Kim YK, Yu H, Persons V, **Donaldson KJ**, Ferdous S, Shelton B, Zhang N, Chrenek MA, Jiang Y, Grossniklaus H, Boatright JH, Kong J, Nickerson JM. Morphometric analysis of retinal pigment epithelial cells from C57BL/6J mice during aging. Investigative Ophthalmology and Visual Science. 2021. 62(2):32. PMID: 33616620

Donaldson KJ, Wu WF, Skelton H, Markand S, Ferdous S, Sellers J, Chrenek MA, Gefke I, Kim SM, Rha J, Liao KL, Grossniklaus HE, Jiang Y, Kong J, Boatright JH, Nickerson JM. Analysis of Damage and Wound Healing in the Retinal Pigmented Epithelium. Advances In Experimental Medicine and Biology: Retinal Degenerative Diseases. 2019. 1185:425-430. PMID: 31884649

Zhang Q, Chrenek MA, Bhatia S, Rashid A, Ferdous S, **Donaldson KJ**, Skelton H, Wu W, See TR, Dalal N, Nickerson JM, Grossniklaus HE. Comparison of histologic findings in age-related macular degeneration with RPE flatmount images. Molecular Vision. 2019. 25:70. PMID: 30820143

Wang J, Struebing FL, Ferdous S, **Donaldson K**, Boatright JH, Geisert EE, Nickerson JM. Differential Exon Expression in a Large Family of Retinal Genes Is Regulated by a Single Trans Locus. In *Retinal Degenerative Diseases* (pp. 413-420). Springer, Cham. 2018. PMID: 29721971

Winkowski D, Nagode D, **Donaldson K**, Yin P, Shamma S, Fritz J, Kanold P. Orbitofrontal cortex neurons respond to sound and activate auditory cortex neurons. Cerebral Cortex. 2017 Jan 9; 1-12. PMID: 29721971

Markand S, Baskin N, Chakraborty R, Landis E, Wetzstein S, **Donaldson K**, Priyadarshani P, Alderson S, Sidhu C, Boatright J, Iuvone P, Pardue M, Nickerson J. IRBP deficiency permits precocious ocular development and myopia. Molecular Vision. 2016 Oct 27; 22:1291-1308. PMID: 27829784

IN REVISION

Shelton DA, **Donaldson K**, Gefke I, Summers V, Kim YK, Yu HY, Getz Y, Ferdous S, Liao K, Papania JT, Chrenek MA, Boatright JH, Nickerson JM. Age-Related RPE changes in Wildtype C57BL/6J Mice between 2 and 32 Months. (Submitted to Molecular Vision, Feb 2025)

PREPRINT

Donaldson KJ, Chrenek MA, Boatright JH, Nickerson JN. High resolution imaging and interpretation of three-dimensional RPE sheet structure. (Submitted to Biomolecules, Feb 2025)
bioRxiv 2024.12.04.626881; doi: https://doi.org/10.1101/2024.12.04.626881

Shelton DA, Gefke I, Summers V, Kim YK, Yu HY, Getz Y, Ferdous S, **Donaldson K**, Liao K, Papania JT, Chrenek MA, Boatright JH, Nickerson JM. Age-Related RPE changes in Wildtype C57BL/6J Mice between 2 and 32 Months. bioRxiv 2024.01.30.574142; doi: https://doi.org/10.1101/2024.01.30.574142

IN PREPARATION

Donaldson KJ, Cunningham C, Lataillade SM, Patel AA, Cox DN. Experience-dependent behavioral, physiological and structural plasticity of cold nociception. (*Manuscript in preparation for submission to Cell Press/Current Biology based on KJD thesis work*)

Donaldson KJ, Papania JT, Gefke I, Summers V, Shelton D, Boatright JH, Nickerson JM. A novel, scalable and accessible, high-throughput pipeline for full murine RPE sheet segmentation and morphometric analysis

Donaldson KJ, Zhang N, Gefke IF, Summers VR, Boatright J, Nickerson J. Alpha catenin dynamics in multiple damage models of the retinal pigment epithelium.

Donaldson KJ, Gefke IF, Summers VR, Getz Y, Skelton H, Nickerson J. *In vivo*, non-invasive longitudinal tracking of RPE dysmorphology in multiple murine damage models.

ORAL PRESENTATIONS

INIVITED

Donaldson KJ. Advanced imaging methods reveal hidden injury and altered alpha-catenin organization in multiple damage models of RPE. Atlanta Vision Research Community Data Session. Emory University. Jul 29, 2025. Atlanta, GA.

CONFERENCE

Donaldson KJ, Skelton HM, Sellers JT, Grossniklaus H, Nickerson JM. Novel *ex* and *in vivo* methods for non-invasive longitudinal tracking of RPE dysmorphology. The Association for Research in Vision and Ophthalmology Annual Meeting; 2018 Apr 29-May 3. Honolulu, HI.

OTHER

Donaldson KJ. Harnessing machine learning and AI to improve neuroscience analysis workflows. Data Analysis Seminar, Neuroscience Institute Georgia State University. Feb 16, 2024. Atlanta, GA**.**

Donaldson KJ. Experience-dependent plasticity and critical periods in an early-life and chronic pain model. Graduate Conference for Research, Georgia State University. Oct 6, 2023. Atlanta, GA.

Donaldson KJ. Experience-dependent plasticity of cold nociception. GSU Neuroscience Institute Breakfast Lecture (NIBL). Mar 25, 2022. Atlanta, GA.

Donaldson, KJ. Visualizing abnormal retinal pigment epithelium. Atlanta Vision Research Community Data Session. Emory University. Nov 7, 2017. Atlanta, GA.

Donaldson KJ. Pupillary abnormality in the interphotoreceptor retinoid binding protein (IRBP) knockout mouse. Atlanta Vision Research Community Data Session. Emory University. Jan 26, 2016. Atlanta, GA.

POSTER PRESENTATIONS

Giradot PE, Zhang X, Zhang N, **Donaldson KJ**, Chrenek MA, Sellers JT, Feola AJ, Nickerson JM, Jain N, Boatright JH. Strain differences in mouse models of pentosan polysulfate retinopathy. Poster presented at: The Association for Research in Vision and Ophthalmology Annual Meeting: Vision For the Future; 2024 May 5-9. Seattle, WA.

Donaldson KJ, Lataillade SM, Williams C, Cunningham C, Patel AA, Cox DN. Determination of Behavioral and Morphological Critical Periods in the Cold Nociceptive System. 52nd Annual Meeting of the Society for Neuroscience; 2023 Nov 11-15. Washington DC.

Donaldson KJ, Cunningham CH, Cox DN. Quantification of nociceptive behaviors using markerless pose estimation and deep learning (DeepLabCut). Presented at: Brains and Behavior Retreat; May 18, 2023. Georgia State University, Atlanta, GA.

Donaldson KJ, Sakurai A, Patel AA, Cox DN. Prior cold experience alters cold-evoked electrophysiological responses and calcium dynamics in cold nociceptors**.** Neuroscience 2022. 51st Annual Meeting of the Society for Neuroscience; 2022 Nov 12-16. San Diego, CA.

Donaldson KJ, Sakurai A, Patel AA, Cox DN. Chronic activation of primary cold nociceptors causes increased behavioral and physiological sensitivity to subsequent thermal stimuli. Presented at: What's New in Atlanta Neuroscience (Atlanta Chapter of the Society for Neuroscience); Nov 3, 2022.

Cunningham C, **Donaldson KJ**, Cox DN. Developmental exposure to varying thermal environments alters noxious cold-evoked behavioral output in *Drosophila* larvae. Poster presented at: Going the Distance Research Symposium. Aug 3, 2022. Atlanta, GA.

Lataillade S, **Donaldson KJ**, Cox DN. Identifying the critical period for cold nociceptive behavioral plasticity in *Drosophila*. Poster presented at: Going the Distance Research Symposium. Aug 3, 2022. Atlanta, GA.

Donaldson KJ, Papania JT, Singh S, Sakurai A, Cox DN. Chronic activation of primary cold nociceptors causes increased sensitivity to subsequent cold challenge. Presented at: Brains and Behavior Retreat; May 12, 2022. Georgia State University, Atlanta, GA.

Giradot PE, Zhang X, Zhang N, **Donaldson KJ**, Chrenek MA, Sellers JT, Jain N, Nickerson JM, Boatright JH. Pentosan polysulfate retinopathy in mice involves subtle changes in outer segment length and RPE cell size. Poster presented at: The Association for Research in Vision and Ophthalmology Annual Meeting; 2022 May 1-4. Denver, CO.

Donaldson KJ, Papania JT, Cox DN. Experience-dependent plasticity of cold nociception in *Drosophila* larvae. Neuroscience 2021. 50th Annual Meeting of the Society for Neuroscience; 2021 Nov 13-17. Chicago, IL.

Donaldson KJ, Papania J, Sakurai A, Himmel NJ, Cox DN. Experience-Dependent Plasticity of Cold-Evoked Nociception. Presented at: Brains and Behavior Retreat; May 20, 2021. Georgia State University, Atlanta, GA.

Himmel NJ, Letcher JM, Sakurai A, Gray TR, Benson MN, **Donaldson KJ**, Cox DN. The evolution of cold nociception in drosopholid larvae and identification of a neural basis for cold acclimation. Presented at: Brains and Behavior Retreat; May 20, 2021. Georgia State University, Atlanta, GA.

Gefke IF, Summers VR, **Donaldson KJ**, Shelton D, Boatright JH, Nickerson JM, Zhang N. Two methods for measuring cytosolic alpha catenin in individually segmented cells of entire retinal pigment epithelium flatmounts. Poster presented at: The Association for Research in Vision and Ophthalmology Annual Meeting; 2021 May 1-7.

Summers, VR, Ferdous S, Liao K, **Donaldson KJ**, Gefke IF, Kim YK, Shelton D, Sellers JT, Markand S, Boatright JH, Nickerson JM. Age-Related Ocular Changes in the RPE of C57/BL6J Mice. The Association for Research in Vision and Ophthalmology Annual Meeting; 2021 May 1-7.

Summers VR, Gefke IF, **Donaldson KJ**, Wu W, Shelton D, Boatright JH, Nickerson JM. A machine-learning based, high-throughput method to quantify abnormal cellular distribution of alpha-catenin following retinal pigment epithelium damage. The Southeastern Vision Research Conference; 2020 Dec 7-8, Atlanta, GA.

Kim Y, Summers VR, **Donaldson KJ**, Ferdous S, Shelton B, Gefke I, Chrenek M, Jiang Y, Kong J, Boatright JH, Nickerson JM. Analysis of RPE cell morphology and numbers from C57/BL6J mice during aging. Poster presented at: The Association for Research in Vision and Ophthalmology Annual Meeting; 2020 May 1-7, Baltimore, MD.

Gefke I, Persons VR, **Donaldson KJ**, Nickerson JM. Patterning of RPE Sheets in Aged Mice. Poster presented at: The Association for Research in Vision and Ophthalmology Annual Meeting; 2020 May 1-7, Baltimore, MD.

Persons V, Ferdous S, Liao K, Wu W, **Donaldson KJ**, Gefke I, Kim Y, Shelton D, Sellers J, Markand S, Boatright JH, Nickerson JM. Age-related ocular changes in C57BL/6J mice from P30-P930. Poster presented at: The Association for Research in Vision and Ophthalmology Annual Meeting; 2020 May 1-7, Baltimore, MD (virtual).

Wu W, **Donaldson KJ**, Liao KL, Gefke I, Boatright JH, Grossniklaus HE, Nickerson JM. Novel *in* and *ex vivo* methods to reveal abnormal RPE morphology and microglial cell migration following light-induced retinal

damage. Poster presented at: From Bench to Bedside and Back: The Association for Research in Vision and Ophthalmology Annual Meeting; 2019 Apr 27-May 2. Vancouver, Canada.

Liao KL, Markand S, Ferdous S, **Donaldson K,** Boatright J, Nickerson JM. The effects of aging on retinal degeneration and myopia in IRBP KO mice. Poster presented at: From Bench to Bedside and Back: The Association for Research in Vision and Ophthalmology Annual Meeting; 2019 Apr 27-May 2. Vancouver, Canada.

Priyadarshani P, Ferdous S, Markand S, **Donaldson KJ**, Boatright JH, Nickerson JM. Characterization of magnetofection transfection reagent, a novel transfection method for gene therapy in a mouse model. Poster presented at: Stand for strong vision science. The Association for Research in Vision and Ophthalmology Annual Meeting; 2018 Apr 29-May 3. Honolulu, HI.

Skelton H, **Donaldson K**, Grossniklaus H, Nickerson J. Morphologic changes in mouse retinal pigment epithelium subsequent to light induced retinal damage. Poster presented at: The 8th Annual Southeastern Medical Scientist Symposium; 2017 Nov 18-19. Atlanta, GA.

Donaldson K, Skelton H, Sellers J, Boatright J, Nickerson J. Novel longitudinal *ex* and *in vivo* methods reveal persistent abnormal RPE morphology following subretinal injections. Poster presented at: The Southeastern Vision Conference; 2017 Oct 2-3; Nashville, TN.

Donaldson K, Sellers J, Boatright J, Nickerson J. Alpha-catenin is a novel marker for identifying abnormal morphology following surgical damage of the RPE. Poster presented at: Global connections in vision research. The Association for Research in Vision and Ophthalmology Annual Meeting; 2017 May 7-11. Baltimore, MD.

Donaldson K, Markand S, Sellers J, Chrenek M, Pardue M, Iuvone P, Boatright J, Nickerson J. Pupillary responses in the C57BL/6J and congenic IRBP knockout mouse. Poster presented at: Research: A vision for hope. The Association for Research in Vision and Ophthalmology Annual Meeting; 2016 Apr 30-May 5. Seattle, WA.

Markand S, Wetzstein S, Williams N, Chakraborty R, **Donaldson K**, Boatright J, Pardue M, Nickerson J. Ocular parameter changes in the IRBP knockout mouse eye. Poster presented at: Research: A vision for hope. The Association for Research in Vision and Ophthalmology Annual Meeting; 2016 Apr 30-May 5. Seattle, WA.

Nickerson J, Struebing F, Markand S, **Donaldson K**, Ferdous S, Boatright J, Geisert E. Alternative splicing in the retina in a large family of genes is regulated by a single trans locus. Paper presented at: Research: A vision for hope. The Association for Research in Vision and Ophthalmology Annual Meeting; 2016 Apr 30-May 5. Seattle, WA.

Priyadarshani P, Guo S, **Donaldson K**, Huang H, Chrenek M, Grossniklaus H, Jiang Y, Nickerson J. Robust automated processing and analysis of images of RPE cells Poster presented at: Research: A vision for hope. The Association for Research in Vision and Ophthalmology Annual Meeting; 2016 Apr 30-May 5. Seattle, WA.

Wetzstein S, Sellers J, **Donaldson K**, Unger G, Markand S, Boatright J, Nickerson J. Intravitreal injection vs. topical eye drop application of nanoparticles using a mouse model. Poster presented at: Research: A vision for hope. The Association for Research in Vision and Ophthalmology Annual Meeting; 2016 Apr 30-May 5. Seattle, WA.

Donaldson K, Yin P, Shamma S, Fritz J. Neuronal responses to acoustic stimuli in the orbitofrontal cortex of the mouse. Poster presented at: 38th Annual MidWinter Meeting of the Association for Research in Otolaryngology; 2015 Feb 21-25; Baltimore, MD.

Fritz J, David S, **Donaldson K**, Shamma, S. Behavioral Gating of Acoustic Information in Ferret Frontal Cortex During Auditory Target Recognition. Poster presented at: Neuroscience 2007. 37th Annual Meeting of the Society for Neuroscience; 2007 Nov 3-7. San Diego, CA.

Fritz J, Elhilali M, Yin P, Harper N, **Donaldson K**, Shamma S. Multiple auditory tasks and the single cortical neuron: salient temporal and spectral cues drive orthogonal, dynamic, task-related receptive field plasticity in primary auditory cortex. Poster presented at: Neuroscience 2005. 35th Annual Meeting of the Society for Neuroscience; 2005 Nov 12-16. Washington, DC.

Christman M, **Donaldson K**, Miller T. Spatial distribution and biodiversity of juvenile fin fish. 132nd Annual Meeting of the American Fisheries Society; 2002. Baltimore, MD.

HONORS AND AWARDS

2024 *Outstanding Graduate Student Mentor*, Neuroscience Institute, Georgia State University

2023 *Kenneth W. and Georganne F. Honeycutt Fellowship*, Neuroscience Institute, Georgia State University
- Merit-based, competitive stipend supplement

2022 *Research to Prevent Blindness Departmental Research Award*, Department of Ophthalmology, Emory University
- Competitive departmental grant ($3000). Title: Identifying the spatial distribution of transcriptionally defined subpopulations of Retinal Pigment Epithelium cells and their differential disease susceptibility

2022-2025 *Brains and Behavior Fellowship*, Georgia State University

2021 *Trainee Professional Development Award*, Leadership Development Program, Society for Neuroscience

2018-2022 *Second Century Initiative (2CI) Neurogenomics Fellowship*, Georgia State University

2018 *Knights Templar Eye Foundation Travel Grant*, ARVO Foundation for Eye Research

2017 *Research to Prevent Blindness Departmental Research Award*, Department of Ophthalmology, Emory University
- Title: Non-invasive longitudinal tracking of cell-mediated inflammatory response dynamics in the living mouse

2016 *Research to Prevent Blindness Departmental Research Award*, Department of Ophthalmology, Emory University
- Title: Development of murine pupilometer device to investigate abnormal pupillary light responses in the IRBP knock-out mouse

2008-2010 *Graduate Research Fellowship,* National Science Foundation – Integrative Graduate Education and Research Traineeship (NSF-IGERT) program for the Biological and Computational Foundations of Language Diversity

TEACHING AND MENTORSHIP

GEORGIA STATE UNIVERSITY

Graduate Teaching Assistant

Spring 2024 Sensory Neuroscience, NEUR4070
- Assisted instructor of record in teaching undergraduate students tenets of sensory neuroscience. Responsible for grading and feedback on weekly assessments.

Guest Lecturer (Graduate Level Course)

Spring 2022 Topics in Neuroscience: Sensation and Perception, NEUR8790

- Invited Lecturer for two lectures to graduate students covering 1) the neural basis and anatomy of retinal processing and sub-cortical vision; 2) peripheral and central auditory sensation and perception

Instructor of Record (Graduate Level Course)

Fall 2019 R Programming Basics of Next Generation Sequencing and Microarray Analysis, NEUR8790

- Created, developed, and taught a curriculum covering bioinformatics analyses using the R programming language to a class of 12 graduate students

Guest Lecturer

Spring 2019 Principles of Neuroscience, NEUR3000

- Lectured on the neural basis of vision, audition, and perception to a class of 40 undergraduate Neuroscience majors.

UNIVERSITY OF MARYLAND, COLLEGE PARK

Guest Lecturer

Spring, Fall 2014 Mammalian Physiology Lecture, BSCI440

- Gave multiple lectures each semester on cardiovascular and pulmonary systems, pharmacology, and neuroscience to class of 100-plus undergraduates

Graduate Teaching Assistant

2010-2014 Mammalian Physiology Laboratory, BSCI441 (8 semesters)

- Taught senior undergraduate students tenets of experimental design, data acquisition, and analysis
- Demonstrated rodent surgical preparation involving catheterization of the carotid artery and jugular vein for subsequent *in vivo* monitoring of physiological responses to pharmacological agents
- Evaluated design, execution, and presentation of student independent projects

Summer 2012-14 Human Anatomy and Physiology I, BSCI201 (3 semesters)

- Taught human anatomical terminology and cellular systems
- Wrote, administered, and graded daily quizzes and lab practicals
- Set up and prepared experiments, anatomical models, and organ dissections

Summer 2012, 2014 Human Anatomy and Physiology II, BSCI202 (2 semesters)

- Taught human physiology concepts and organ systems
- Responsibilities consistent with BSCI201

Summer 2011 Principles of Biology I Laboratory, BSCI105 (1 semester)

- Instructed junior undergraduate students in applying the Scientific Method to basic questions in cellular and molecular biology via lab exercises
- Prepared requisite materials and maintained equipment for molecular biology assays

Fall 2007 Neurophysiology Laboratory, BSCI454 (1 semester)

- Instructed senior undergraduate students in experimental design and electrophysiological techniques using crayfish and roach models
- Maintained and prepared multiple electrophysiology rigs for weekly experiments

MENTORSHIP

2021-2025 *Georgia State University Neuroscience Institute – 1st year PhD students*

- Peer mentoring of incoming first-year graduate students as part of Neuro Buddy Program, incorporating DEI and Cultural Competency training

2020-2023 *Cox Lab - Undergraduate Research Assistants*

- Trained and mentored 4 Georgia State undergraduate students
  - Sabria Lataillade: Jan 2022 - Dec 2022
    - Beckman Scholar, GSU Brains and Behavior Summer Fellow 2022
    - Completed Honors Thesis: Identifying a critical period for cold nociceptive behavioral plasticity
  - Cole Cunningham: Nov 2021-Dec 2023
    - GSU Brains and Behavior Summer Fellow 2022
    - Completed Honors Thesis: The effect of differential thermal rearing on cold nociception
  - Sumeet Singh: June-Oct 2021
  - Jack Papania: Sep-Dec 2020

2017-2018 *Nickerson Lab – Emory School of Medicine medical student research trainees*

- Mentored 2 medical student volunteers to create and carry out independent research projects related to RPE damage

Spring 2017 *Georgia Tech - Senior Capstone Project*

- Mentored 4 engineering students through the processes of designing, prototyping, and testing of a novel mouse pupilometer
- Prototype presented at Georgia Tech Capstone Design Expo, attended by the general public, and featured on local Atlanta news stations

2010-2014 *UMCP Neural Systems Lab - Undergraduate Assistants*

- Mentored 12 undergraduate research assistants to carry out mouse behavioral training and histological experiments
- Two upperclassmen submitted senior thesis projects and graduated with honors
- Nine students pursued professional training (6 medical school, 1 dental school, 2 graduate school) immediately following graduation. Two received NIH Postbaccalaureate Intramural Research Training Awards

SERVICE AND COMMITTEES

2024	GSU Symposium for Undergraduate Neuroscience Poster Judge
2024	NI Breakfast Lecture Workshop on Scientific Presenting Panelist
2024	Neuroscience Methods Journal Club (GSU NI) Co-Founder
2023	Atlanta Science Festival Neuroscience Exhibit Coordinator: Bringing Optogenetics to the People
2022	Neuroscience Institute Seminar Speaker Selection Committee member
2022	Atlanta Science Festival Neuroscience Exhibitor: Bringing Optogenetics to the People
2021	Atlanta Regional Brain Bee Chair
2020	Atlanta Regional Brain Bee Workshop Instructor and Emcee/Host
2019	Atlanta Regional Brain Bee Workshop Instructor
2018-2023	Georgia State University, Neuroscience Institute outreach committee member
2013	Neuroscience and Cognitive Science Graduate Program (NACS) outreach committee member
2011-2013	UMCP Maryland Day Campus Open House - NACS/Linguistics outreach volunteer
2010	IGERT outreach committee, High School Visitation lecturer
2009	IGERT program external review volunteer/liaison
2008-2009	NACS Student host and discussion leader for weekly Invited Lecturer Seminar Series

PROFESSIONAL SOCIETY MEMBERSHIPS

2018-Present	American Association for the Advancement of Science
2015-2018	Association for Research in Vision and Ophthalmology
2015	Association for Research in Otolaryngology
2005, 2007, 2021-2025	Society for Neuroscience

ADDITIONAL TRAINING

2022	Summer Data Jam: Analyzing, Visualizing & Communicating for Maximum Impact (GRAD7999) *Georgia State University Graduate School*
2020	Advanced Sequencing Technologies & Bioinformatics Analysis (SEQTEC2020) *Cold Spring Harbor Laboratory*